



Compost Nutrient Balance Analysis Results and Water Quality Implications from a Demonstration Livestock Manure Windrow Composting Site

David Webber, Heekwon Ahn, Thomas Richard & Steven Mickelson

To cite this article: David Webber, Heekwon Ahn, Thomas Richard & Steven Mickelson (27 Mar 2025): Compost Nutrient Balance Analysis Results and Water Quality Implications from a Demonstration Livestock Manure Windrow Composting Site, Compost Science & Utilization, DOI: [10.1080/1065657X.2025.2476395](https://doi.org/10.1080/1065657X.2025.2476395)

To link to this article: <https://doi.org/10.1080/1065657X.2025.2476395>



Published online: 27 Mar 2025.



Submit your article to this journal [↗](#)



View related articles [↗](#)



View Crossmark data [↗](#)



Compost Nutrient Balance Analysis Results and Water Quality Implications from a Demonstration Livestock Manure Windrow Composting Site

David Webber^a, Heekwon Ahn^b, Thomas Richard^c and Steven Mickelson^a

^aDepartment of Agricultural and Biosystems Engineering, Iowa State University, Ames, Iowa; ^bDepartment of Animal Biosystem Science, Chungnam National University, Daejeon, South Korea; ^cDepartment of Agricultural & Biological Engineering, Pennsylvania State University, University Park, Pennsylvania

ABSTRACT

Windrow composting is an effective treatment method for livestock manure. However, few studies have investigated the detailed relationship between the composting process and surface water quality impacts on surrounding areas. This study conducted a compost nutrient balance analysis (CNBA) of composting materials under natural field conditions using livestock manure samples collected at a research windrow composting site in central Iowa, USA. The research site included a fly ash composting pad surface and vegetative filter strip (VFS) buffer plots. Surface runoff and pollutant transport data that included sediment (solids), nitrogen (N), and phosphorus (P) nutrient concentrations and losses also were collected and analyzed. The CNBA results indicated that all composting properties and materials mostly followed the general compost biodegradation-related trends throughout spring (early) and summer (late) 60-day composting periods. However, the results also showed that while an average of 34% P was lost due to the composting process, only an average of 0.3% P was lost to runoff from the no-VFS buffer (control) plots during the composting periods. Published research indicates significantly lower P losses from control plot runoff may be attributed to the chemical and physical conversion effects of the P-sorbent fly ash pad surface material. Consequently, further research should be conducted that considers the use of fly ash or other P-sorbent pad surface materials in the construction of new windrow composting sites and the upgrading of existing composting facilities.

Introduction

Livestock manure management activities have resulted in serious environmental pollution issues for air, land, and water resources (Ramos, Quinton, and Tyrrel 2006; Smith et al. 2007; Burkholder et al. 2007; James et al. 2007). Agricultural nutrients in the form of nonpoint-source (NPS) pollution continue to be a significant contributor to poor water quality in the U.S. as determined by USEPA (2009) and USGS (2010). These NPS pollutants also have been reported as contributors to hypoxic conditions in the Gulf of Mexico by David, Drinkwater, and McIsaac (2010) and Jacobson, David, and Drinkwater (2011).

Livestock manure can provide an excellent source of organic matter and plant nutrients for various cropping system fertilization inputs.

Windrow composting is one strategy to minimize the adverse effects of livestock manure on the environment and also provide a soil amendment product for enhancing crop fertilizer applications. Surface runoff concentrations of nutrients such as carbon (C), nitrogen (N), phosphorus (P), and potassium (K) also can be elevated by land application of manure to agricultural fields (Edwards and Daniel 1993; Heathwaite, Griffiths, and Parkinson 1998; Burton and Turner 2003; Ahmed et al. 2013; Webber et al. 2009). Parry (1998) found that nutrient runoff from agricultural fields is a major source of water pollution in surface waters. However, negative impacts from composting facilities due to surface runoff also can occur even under proper composting site management and normal weather conditions (Webber et al. 2016).

Research has shown that composted manure is less hazardous to the environment versus raw manure (Eghball and Power 1999; Vervoort et al. 1998) because a higher fraction of total N is in stable organic forms (Rynk et al. 1992; Tiquia, Richard, and Honeyman 2002). The compost-amended soils have also been shown to significantly reduce nitrate-N ($\text{NO}_3\text{-N}$) leaching relative to conventional fertilizers (Maynard 1993). However, research findings and related information compiled by Webber et al. (2016) show that a major disadvantage of windrow composting is nutrient loss during the composting process, which can occur through leaching, runoff, and volatilization. Results of one study reported by Wilson, Haralampides, and Levesque (2004) reflecting the potential severity of nutrient losses from saturated compost windrows indicated that approximately 68% of both natural and simulated rainfall resulted in runoff. Windrow composting site surface runoff water quality also can be affected by compost maturity (Bernal, Albuquerque, and Moral 2009; Larney et al. 2014a, 2014b).

Compost nutrient balance analysis (CNBA) and related quantitative studies have been conducted by several researchers (Robinson, Kimmel, and Avnimelech 2000; Barrington et al. 2002; Hansen, Henriksen, and Sommer 2006; Farran et al. 2006; Ahn, Richard, and Choi 2007; Andersen et al. 2010, 2011; Breitenbeck and Schellinger 2004; Fan et al. 2019; Sefeedpari et al. 2019). Tiquia, Richard, and Honeyman (2002) determined there were 20% to 60% losses of N, P, and K during the composting process. These losses can occur in the gas or liquid phase through volatilization, runoff, and leachate infiltration, as reported by Tiquia, Richard, and Honeyman (2002) and Webber et al. (2016). Seymour and Bourdon (2003) also found that concentrations of $\text{NO}_3\text{-N}$, ortho-P ($\text{PO}_4\text{-P}$), and K were highest in leachate compared to runoff from compost windrows under natural rainfall conditions.

Lam, Schmalz, and Fohrer (2011) determined that a combination of best management practices (BMPs) tended to be most effective in improving water quality. A combination of BMPs could include the industrial by-product fly ash that was

used for the windrow composting pad surface on which these CNBA and surface runoff studies were conducted. Fly ash that has been properly prepared and installed at a windrow composting site has been reported to function as a nearly impervious surface material for windrow composting pad construction (Sikora and Francis 2000). A potential water quality improvement benefit of the fly ash pad surface may stem from the P-sorption properties of the fly ash material reported by Dou et al. (2003).

Few studies in the scientific literature have investigated the relationship between the compost biodegradation process sediment and nutrient mass losses versus these losses *via* surface runoff from windrow composting sites. The primary objective of this investigation was to estimate compost sediment and nutrient mass losses due to biodegradation of organic matter during the windrow composting process. This study also endeavored to analyze and compare these data with surface runoff water quality data and the reported nutrient conserving efficacy data of the P-sorbent fly ash windrow composting pad surface material. Consequently, further research should be conducted that considers the use of fly ash or other P-sorbent pad surface materials in the construction of new windrow composting sites and the upgrading of existing composting facilities.

Hypothesis

Nutrient losses from compost windrows can only occur through the compost biodegradation process, leaching, volatilization, and water-borne surface runoff from the composting pad surface.

Materials and Methods

Windrow Composting Research Site

The study site was located at the former Iowa State University Dairy Teaching Farm, Ames, central Iowa, USA ($42^\circ 0' 34''$ N, $93^\circ 39' 16''$ W). The study site was 0.25 ha and comprised a total area of nine plots, each 6.0 m-wide $\times$ 46 m-long. The research plot area was selected on uneven terrain with an average slope of 5%, with each of the six VFS buffer plots and three no-VFS

buffer (control) plots equivalent to 6.0m-wide × 23m-long. The 1:1-, 1:0.5-, and 1:0-area ratio

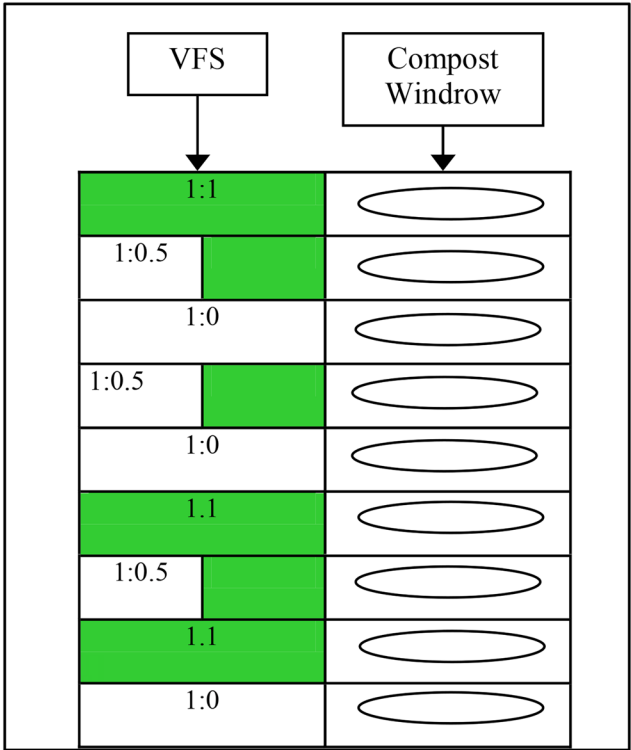


Figure 1. Windrow composting piles and research plot layout scheme.

plot treatments referred to the ratio of the contributing area: full-length area, half-length area, and no-buffer (control) plots, respectively. Each treatment included three replications for a total of nine plots (Figure 1). The CNBA results in this paper are based only on the three 1:0-area ratio control plots.

The upslope windrow composting pad surface area was divided into nine 6.0m-wide × 23m-long plots that each contained one of the nine compost windrows. The plots included linear surface barriers between each compost windrow to prevent cross-contamination of surface water in the tipping-bucket runoff sampling units (Hansen and Goyal 2001). The protective barriers comprised 25m-long fire hose sections filled with water that could be drained for easy removal and then re-filled and replaced to expedite the compost windrow maintenance activities (Figure 2).

The pad surface was constructed of fly ash, a by-product of combustion from coal-fired power plants provided by Alliant Energy, Inc., Marshalltown, Iowa, USA. The 0.13ha composting pad area was prepared by machine grading to approximately a 2% average slope, and fly ash was hydro-compacted to an average depth of 31 cm.



Figure 2. Compost windrows and runoff collection system. The vinyl fire hoses provided mobile barriers between the windrow plots that were easily drained and moved for compost turning and sampling operations.

Compost Nutrient Balance Analysis Procedures

This study was conducted concurrently with an investigation of windrow composting site nutrient concentrations and total losses in surface runoff (Webber et al. 2009). The composting period was based on a 60-day duration during a particular research season, occurring approximately during June and July (early season [ES]) and August and September (late season [LS]) time periods (Table 1). Dairy cow manure (2002LS and 2003ES), horse manure (2004ES), and a dairy cow/horse/sheep manure mixture (2004LS) were composted in full-scale windrow systems, each trial containing nine windrows. Samples were collected three times during a composting period at 0, 30, and 60 days after compost windrow construction. A total of three, waist-height (1.0 m-high) “grab-sample” runs per windrow were conducted, with a collection of about a 3.0 L-volume sample per run for nutrient analysis. Compost windrows were turned with tractor-assisted, elevating-face conveyor and rotary

drum flail-type compost turning implements every week for the first two weeks and bi-weekly for the remainder of the 60-day composting period.

The initial stage of the CNBA study included documenting the physical compost characteristics by identifying the manure type and determining the initial manure weight (kg), and percent (%) initial moisture content (MC) and volatile solids (VS) values (Table 1). The next stage involved conducting further CNBA procedures using compost nutrient data from Table 1 (percent [%] stone content [SC], total carbon [TC], total nitrogen [TN], and carbon/nitrogen [C/N] ratio, and mass concentration [mg kg^{-1}] values of total phosphorus [TP] and total potassium [TK]). The data in Tables 2 and 3 indicate the change in total mass of VS and TC, and TN, TP, and TK, respectively.

Figures 3–10 provide statistically-based estimated trend variations ($p < 0.05$) for all physical compost characteristics and nutrient contents due

Table 1. Rainfall event depths and initial (day 0) raw manure (type) and average weight and content characteristics including moisture content (MC), stone content (SC), volatile solids (VS), total carbon (TC), total nitrogen (TN), total phosphorus (TP), total potassium (TK), and carbon/nitrogen (C/N) ratios determined from windrow composting plots.

	2002	2003	2004	
	LS ^a	ES ^b	ES	LS
Rainfall depth (mm)	35	142	46	79
Manure (type)	Dairy cow	Dairy cow	Horse	Dairy/Horse/Sheep
Weight ^c (kg, wb ^d)	24,598 ± 1,254 ^a	16,012 ± 1,816	11,661 ± 2,227	8,498 ± 699
MC ^e (% wb)	72 ± 2.9	72 ± 2.1	61 ± 2.1	62 ± 3.0
SC ^e (% wb)	19 ± 3.4	20 ± 1.7	12 ± 3.0	12 ± 3.4
VS ^e (% db ^f)	46 ± 5.3	42 ± 2.2	51 ± 4.5	54 ± 3.1
TC ^e (% db)	19 ± 2.0	19 ± 2.1	20 ± 2.1	28 ± 3.1
TN ^e (% db)	1.9 ± 0.3	1.6 ± 0.1	1.4 ± 0.1	2.0 ± 0.3
C/N ^e (ratio)	9.3 ± 0.9	12 ± 0.9	14 ± 0.7	14 ± 2.0
TP ^e (mg kg ⁻¹ , db)	4,579 ± 531	6,011 ± 1067	5,485 ± 986	7,176 ± 1605
TK ^e (mg kg ⁻¹ , db)	14,914 ± 2247	26,840 ± 2930	24,514 ± 2416	14,631 ± 2365

^aLate season, ^bEarly season, ^cValues are means from nine piles, ^dWet-based, ^eValues are means from 27 samples (three samples from nine piles), ^fDry-based, ^g± One standard deviation.

Table 2. Change in total mass values of volatile solids (VS) and total carbon (TC) due to compost biodegradation during the composting process.

		2002	2003	2004	
		LS ^a	ES ^b	ES	LS
VS (kg)	Initial mass (kg)	2,461 ± 129 ^c	1,944 ± 242	1,512 ± 339	1,471 ± 199
	Final mass (kg)	1,085 ± 175	1,372 ± 150	1,018 ± 240	1,202 ± 76
	Reduced mass (kg)	1,377 ± 206	571 ± 178	590 ± 270	268 ± 173
	% of reduction	56 ± 7	29 ± 7	36 ± 17	17 ± 9
	<i>p</i> -value	<0.001	<0.001	0.004	0.013
TC (kg)	Initial mass (kg)	1,058 ± 75	883 ± 128	636 ± 116	744 ± 128
	Final mass (kg)	431 ± 79	494 ± 33	368 ± 116	554 ± 45
	Reduced mass (kg)	627 ± 100	390 ± 129	312 ± 120	190 ± 95
	% of reduction	59 ± 7	43 ± 8	47 ± 15	24 ± 10
	<i>p</i> -value	<0.001	<0.001	0.002	0.005

^aLate season, ^bEarly season, ^c± One standard deviation.

to compost biodegradation throughout all composting periods. The final stage of the research process dealt with comparing CNBA results to surface runoff water quantity and quality data to determine the fate of compost solids and nutrient mass losses *via* surface runoff from the 1:0-area ratio no-buffer (control) compost windrow plots shown in Table 4. A tipping-bucket flow meter system (Hansen and Goyal 2001) was used to measure and collect runoff samples from each plot within one day following a rainfall event.

The CNBA procedures involved preparing, analyzing, and quantifying composting process conditions, materials, and nutrient data from compost samples collected from the nine windrow composting plots. The MC was determined by oven drying at 105°C for 24h and calculated using the percentage of MC based on the wet-sample analysis (Ahn et al. 2009). The VS values were determined from igniting the dried compost samples in a muffle furnace at 550°C for 24h and calculating the VS content on a dry-sample percentage basis (Ahn et al. 2009).

The SC was estimated by washing the remaining ash after VS analysis. Water was poured into the remaining ash after VS analysis and then shaken to remove the floating ash. This process was repeated several times until no more ash floated to the surface. The presence of SC in compost samples is primarily due to its occurrence in livestock manure and straw bedding materials, and from gravel materials picked up during compost windrow construction, turning, and other maintenance activities. Consequently, the estimation of SC is an essential

process since it is required to correctly convert quantity values in the CNBA calculations.

The nutrient analysis component of the CNBA process involved determining the percentage-based concentration values of TC and TN by detection using a Perkin-Elmer P2400 Elemental Analyzer (Perkin-Elmer Inc., 2004). The analysis of TP and TK initially involved subjecting the samples to microwave digestion in a CEM-MDS 2000 Microwave Digester (CEM Corp., 2006). Subsequently, the elemental percentage-based concentrations of TP and TK were measured using a Perkin-Elmer AANALYST-100 atomic absorption spectrophotometer (Perkin-Elmer Inc., 2004) and finally converted to mass concentration (mg kg⁻¹) and subsequently total mass losses (kg) values.

The Statistical Analysis System (SAS Institute 2004) software was used to analyze and compare

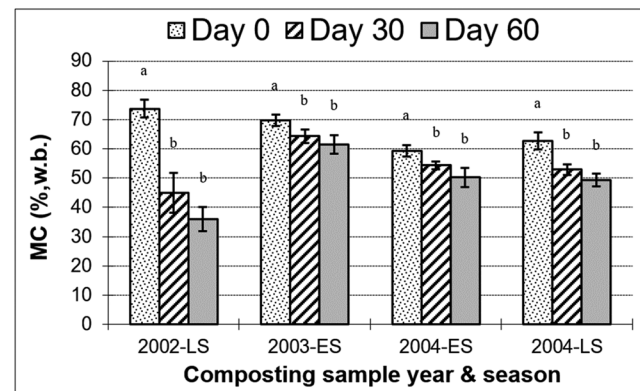


Figure 3. Variation of moisture contents (MC) during composting. Error bars represent one standard deviation, and a different letter (b) indicates significant differences at the 95% probability level ($p < 0.05$).

Table 3. Change in total nutrient mass values of total nitrogen (TN), total phosphorus (TP), and total potassium (TK) due to compost biodegradation during the composting process.

		2002	2003	2004	
		LS ^a	ES ^b	ES	LS
TN (kg)	Initial mass (kg)	114 ± 8 ^c	72 ± 8	44 ± 9	53 ± 8
	Final mass (kg)	38 ± 7	42 ± 5	25 ± 6	45 ± 3
	Reduced mass (kg)	75 ± 10	30 ± 9	19 ± 10	8 ± 6
	% of reduction	66 ± 6	41 ± 9	40 ± 22	13 ± 11
	p-value	<0.001	<0.001	<0.001	0.028
TP (kg)	Initial mass (kg)	26 ± 2	25 ± 4	16 ± 4	18 ± 4
	Final mass (kg)	13 ± 2	12 ± 2	12 ± 3	18 ± 3
	Reduced mass (kg)	13 ± 4	14 ± 5	4 ± 4	0 ± 3
	% of reduction	49 ± 11	53 ± 11	21 ± 32	0 ± 6
	p-value	<0.001	<0.001	0.019	0.743
TK (kg)	Initial mass (kg)	90 ± 8	128 ± 17	78 ± 15	40 ± 5
	Final mass (kg)	37 ± 8	58 ± 6	24 ± 7	33 ± 2
	Reduced mass (kg)	53 ± 10	70 ± 19	54 ± 16	7 ± 3
	% of reduction	59 ± 9	54 ± 8	68 ± 13	16 ± 7
	p-value	<0.001	<0.001	<0.001	0.004

^aLate season, ^bEarly season, ^c± One standard deviation.

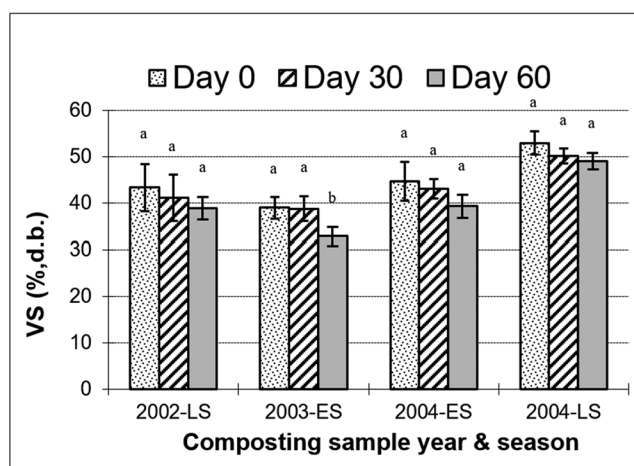


Figure 4. Variation of volatile solids (VS) during composting. Error bars represent one standard deviation, and a different letter (b) indicates significant differences at the 95% probability level ($p < 0.05$).

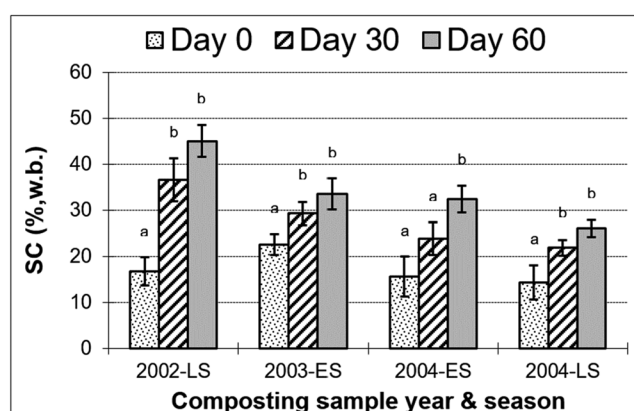


Figure 5. Variation of stone contents (SC) during composting. Error bars represent one standard deviation, and a different letter (b) indicates significant differences at the 95% probability level ($p < 0.05$).

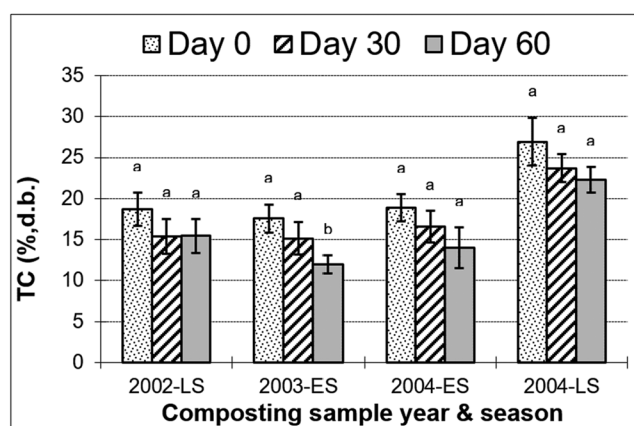


Figure 6. Variation of total carbon (TC) during composting. Error bars represent one standard deviation, and a different letter (b) indicates significant differences at the 95% probability level ($p < 0.05$).

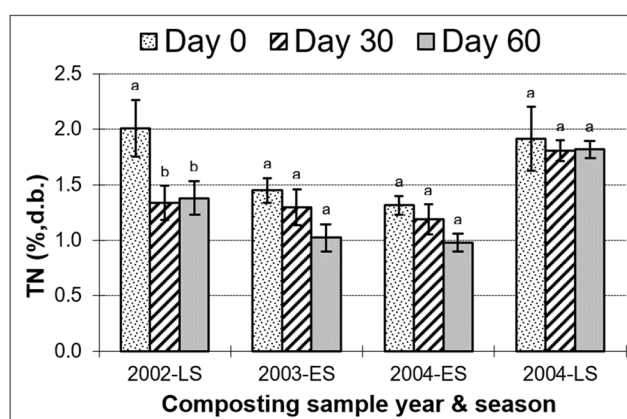


Figure 7. Variation of total nitrogen (TN) during composting. Error bars represent one standard deviation, and a different letter (b) indicates significant differences at the 95% probability level ($p < 0.05$).

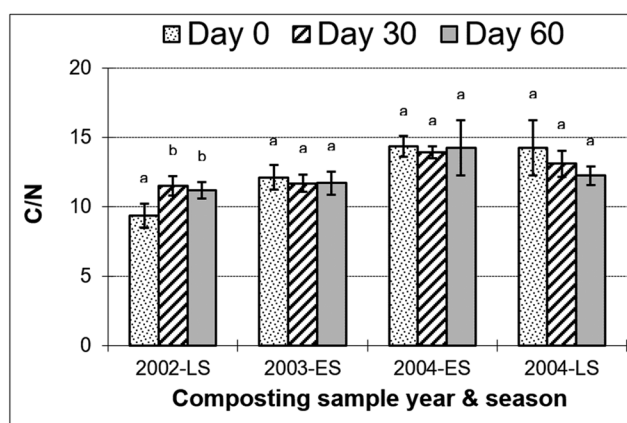


Figure 8. Variation of carbon/nitrogen (C/N) ratio. Error bars represent one standard deviation, and a different letter (b) indicates significant differences at the 95% probability level ($p < 0.05$).

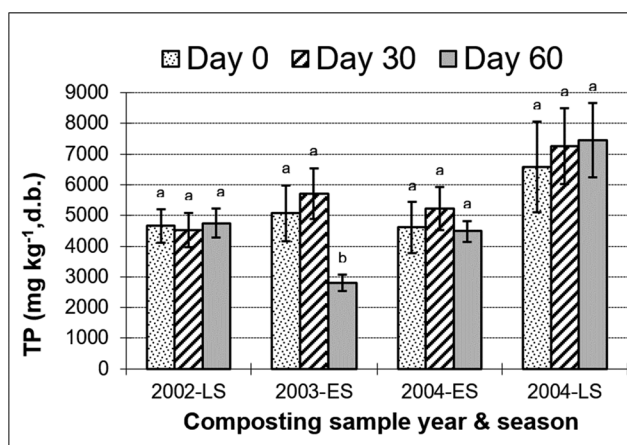


Figure 9. Variation of total phosphorus (TP) mass concentration. Error bars represent one standard deviation, and a different letter (b) indicates significant differences at the 95% probability level ($p < 0.05$).

compost nutrient mass balance and surface runoff analysis data. Significance among treatments was determined using the SAS General Linear Model (GLM) Procedure and Least Squares Mean (LSMEANS) Test. These statistical methods were used to analyze compost windrow plot treatment mean differences among and within the composting period year and season data at the 95% probability level ($p < 0.05$).

Results and Discussion

Variations in Compost Characteristics, Solids, and Nutrients During the Composting Process

Results of the CNBA in Figures 3–10 conducted on samples from the nine compost windrow research plots show estimated 60-day trend variations of compost characteristics, solids, and nutrients during the 2002–2004 ES and LS

composting periods. Figure 3 shows a significant decrease ($p < 0.05$) in percent (%) MC between Day 0 and Day 30 during all composting period years and seasons. This variation in MC is due primarily to heat generated by the biological metabolism and airflow over the compost windrow pile surface, which leads to increased evaporation during the composting process.

Figure 4 shows the variation in %VS, which depicts a pattern of reduction due to biodegradation during all composting periods. However, no significant change ($p < 0.05$) in VS levels occurred during the 2002LS, and 2004ES and 2004LS composting periods. Unlike others, the 2003ES composting period showed significant VS reduction ($p < 0.05$) owing to the relatively large amount of precipitation that is considered to promote decomposition of organic matter in the latter part of the composting process by maintaining favorable moisture conditions for the composting microorganisms.

Figure 5 shows the variation of %SC. Since the windrow compost piles were constructed on the ground and may include various bedding and gravel materials from livestock stalls and the composting pad surface, respectively, these compost materials can contain a significant amount of SC. It is essential to accurately estimate SC in order to correctly convert the analysis results. As the organic compost material biodegraded and MC (Figure 3) was significantly reduced ($p < 0.05$), Figure 5 shows the portion of SC in samples significantly increased ($p < 0.05$) throughout all composting periods.

Figure 6 shows the variation of %TC in the compost samples. Although a reduction in TC is generally observed due to the microbial

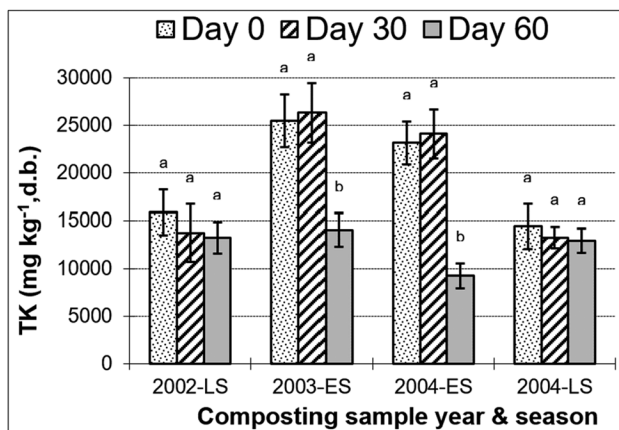


Figure 10. Variation of total potassium (TK) mass concentration. Error bars represent one standard deviation, and a different letter (b) indicates significant differences at the 95% probability level ($p < 0.05$).

Table 4. Total mass values of total solids (TS), nitrate-nitrogen ($\text{NO}_3\text{-N}$), ortho-phosphorus ($\text{PO}_4\text{-P}$), and total phosphorus (TP) due to losses in surface runoff from rainfall event depths among research years and seasons from 1:0 area ratio (control) windrow composting plots. Different lower-case letters denoting comparisons among years (b) and seasons (c), respectively, indicate significant differences at the 95% probability level ($p < 0.05$).

1:0 Area ratio	2002	2003	2004	
	LS ^a	ES ^b	ES	LS
Rainfall depth (mm)	35	142	46	79
TS (kg)	6.4a	12b	0.7a	1.4a
$\text{NO}_3\text{-N}$ (kg)	0.0006a	0.1046b	0.0023a	0.2544b
$\text{PO}_4\text{-P}$ (kg)	0.0105a	0.0156b	0.0004(0.1%) ^c	0.0018(0.4%) ^c
TP (kg)	0.0104a	0.0193b	0.0007c	0.0027c

^aLate season, ^bEarly season, ^cPercent (%) runoff fraction of total $\text{PO}_4\text{-P}$ losses in parenthesis.

degradation of TC for their growth, this reduction was only statistically significant ($p < 0.05$) during the 2003ES composting period. Since the 2003ES composting period had the highest rainfall event total of 142 mm (Table 1), this high-moisture condition may have contributed to the reduction of TC through increased microbial action and leaching of TC from the saturated compost windrows.

Figure 7 indicates the variation of %TN during the composting period. The pathway of TN loss from the compost windrows was probably through emissions of NH_3 and to a lower extent *via* NO_x emissions. Because the 2002LS composting period had the lowest rainfall event total of 35 mm (Table 1), the drier conditions, along with a significantly lower C/N ratio (Figure 8) and a relatively high initial raw manure TN content, may have accelerated TN emissions due to unsaturated compost windrows, lower humidity, and increased evaporation.

Figure 8 represents the variation of the C/N ratio. The initial C/N ratios are low in this experiment. Generally, an initial C/N ratio between 30 and 35 would yield the most efficient composting process (Brito, Coutinho, and Smith 2008). While a higher C/N ratio would lead to a longer composting phase due to higher levels of TC, a lower ratio will result in rapid composting with increasing losses of excess TN from ammonia volatilization (Haug 1993). Although there were no substantial changes indicated in the C/N ratio during the 2003ES and 2004LS composting periods, the significant increase ($p < 0.05$) during the 2002LS composting period may be the result of a significant decrease ($p < 0.05$) in TN shown in Figure 7.

Figure 9 shows the variation of TP mass concentrations (g kg^{-1}). Levels of TP did not significantly change ($p < 0.05$) during the 2002LS and 2004ES and LS composting periods. Although TP is a nonvolatile nutrient that is not generally affected by the composting process, TP can leach out of the compost windrows in runoff. Since the 2003ES composting period had the highest rainfall event total of 142 mm (Table 1), the wet conditions during that composting period may have contributed to the significant reduction ($p < 0.05$) of TP mass concentrations through excessive leaching.

Figure 10 indicates the variation of TK mass concentrations (g kg^{-1}) during the composting

periods. The levels of TK generally decrease during the composting process, mainly due to runoff and leaching. While this estimated pattern was trending in that direction during the 2002LS and 2004LS composting periods, the reduction of TP was significant ($p < 0.05$) during the 2003ES and 2004ES composting periods. This change may have been the result of high moisture conditions during the 2003ES composting periods.

Although the 2004ES composting period had a rainfall event total of only 46 mm, substantially high antecedent moisture conditions that included elevated rainfall/snowfall precipitation of 420 mm during January–May 2004 and rainfall precipitation of 208 mm during May 2004, may have contributed to saturated surface conditions that resulted in significant runoff and leaching of high mass concentration levels of TK (IEM 2018).

Compost Solids and Nutrient Total Mass Losses Due to Compost Biodegradation

Tables 2 and 3 indicate changes in total mass values (kg) of compost solids and nutrients between Initial mass (Day 0) and Final mass (Day 60) sample collection dates *via* compost biodegradation during the composting process. These data show general trends of significant differences ($p < 0.05$) for VS and TC (Table 2) and TN, TP, TK (Table 3) during 2002LS, 2003ES, and 2004ES and LS research seasons. The significant differences ($p < 0.05$) in total nutrient mass values of TC during all 2002–2004 composting years and seasons are described for Figure 6 as a function of the composting-microbial action necessary for growth and the increased leaching effect on TC from saturated compost windrows.

However, Table 2 also shows no significant differences ($p < 0.05$) for VS during the 2004ES and LS seasons, and Table 3 indicates no significant differences ($p < 0.05$) for TN, TP, and TK during the 2004LS research season. These statistically insignificant changes in total mass values of VS, TN, TP, and TK are probably due to the relatively drier conditions during the 2002LS and 2004ES and LS research seasons.

Significantly higher total mass losses ($p < 0.05$) of VS, TC, TP, and TK during the 2003 ES season may be due to a higher rainfall level of

142 mm (Table 1) and the total mass losses mechanisms for these pollutants as described for Figures 4, 6, 9, and 10, respectively. However, the total nutrient mass losses (kg) of TC were significantly different for all composting years and seasons, indicating that although TC total mass losses mechanisms were not substantially affected by compost microbial activity and leaching, estimated variations of percentage-based (%) TC levels (Figure 6) were not significantly different in all but the 2003ES composting period.

Compost Solids and Nutrient Total Mass Losses Due to Surface Runoff

Table 4 shows rainfall depths (mm) and total mass values (kg) of compost solids and nutrients lost due to surface runoff exclusively from the 1:0 area ratio no-VFS buffer (control) windrow composting plots. These data indicate significant differences ($p < 0.05$) in total losses of total solids (TS = compost sediment and other solids in runoff), $\text{NO}_3\text{-N}$, $\text{PO}_4\text{-P}$, and TP during the 2003ES season. These results may be due to the high precipitation and runoff events total of 142 mm (Tables 1 and 4). While the relatively lower rainfall and runoff conditions of 2002LS and 2004ES and LS seasons may have reduced the transport of $\text{PO}_4\text{-P}$ and TP in the runoff stream, there also were significantly lower ($p < 0.05$) runoff losses of $\text{PO}_4\text{-P}$ (0.1% and 0.4%) compared with compost biodegradation mass losses of 41% and 26% (Table 4), respectively.

These nutrient mass balance and comparable surface runoff analysis results indicate $\text{PO}_4\text{-P}$ was removed from the runoff stream in a higher proportion than a similar study where fly ash was not used in composting pad surface construction (Seymour and Bourdon 2003). Results from this CNBA study and other findings also indicated the $\text{PO}_4\text{-P}$ in runoff from the compost windrows may have been absorbed and converted to more stable P compounds by the fly ash composting pad material (Dou et al. 2003).

Conclusions

The compost nutrient balance analysis (CNBA) results indicated that MC, VS, SC, TC, TN, TP,

and TK generally followed the normal compost biodegradation-related trends throughout the 60-day composting process in both early season (ES) and late season (LS) composting periods. These CNBA results also showed significant losses of TN in 2002LS; and of TS, $\text{NO}_3\text{-N}$, $\text{PO}_4\text{-P}$, TP, and TK in 2003ES, were most likely due to dry and wet conditions of the associated composting seasons, respectively. While significant losses of TK occurred during the relatively low-precipitation 2004 ES composting period, high antecedent moisture levels may have contributed to saturated surface conditions that resulted in significant runoff and leaching of total nutrient mass losses of TK.

While 41% and 26% of $\text{PO}_4\text{-P}$ were lost from the compost windrows during 2004ES and LS composting periods due to compost biodegradation, respectively, only 0.1% and 0.4% were lost to runoff from the 1:0 area ratio plots during the respective 2004ES and LS composting periods. The significantly lower $\text{PO}_4\text{-P}$ and TP losses in runoff from the control plots reported in this CNBA study and based on other published results (Sikora and Francis 2000; Dou et al. 2003) may be attributed to the chemical and physical absorption and conversion effects of the fly ash composting pad surface material. Therefore, the potential use of a fly ash or other P-sorbent-type composting pad surface material should be further researched to determine if these materials should be recommended when upgrading current windrow composting facilities and in the construction of new composting facilities.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

References

- Ahmed, S. I., S. K. Mickelson, C. H. Pederson, J. L. Baker, R. S. Kanwar, J. C. Lorimor, and D. Webber. 2013. "Swine Manure Rate, Timing, and Application Method Effects on Post-Harvest Soil Nutrients, Crop Yield, and Potential Water Quality Implications in a Corn-Soybean Rotation." *Transactions of the American Society of Agricultural and Biosystems Engineers* 56 (2): 395–408. doi:10.13031/2013.42678.
- Ahn, H. K., T. J. Sauer, T. L. Richard, and T. D. Glanville. 2009. "Determination of Thermal Properties of

- Composting Bulking Materials.” *Bioresource Technology* 100 (17): 3974–3981. doi:10.1016/j.biortech.2008.11.056.
- Ahn, H. K., T. L. Richard, and H. L. Choi. 2007. “Mass and Thermal Balance during Composting of a Poultry Manure—Wood Shavings Mixture at Different Aeration Rates.” *Process Biochemistry* 42 (2): 215–223. doi:10.1016/j.procbio.2006.08.005.
- Andersen, J. K., A. Boldrin, T. H. Christensen, and C. Scheutz. 2010. “Mass Balances and Life-Cycle Inventory for a Garden Waste Windrow Composting Plant.” *Waste Management & Research: The Journal of the International Solid Wastes and Public Cleansing Association, ISWA* 28 (11): 1010–1020. doi:10.1177/0734242X09360216.
- Andersen, J. K., A. Boldrin, T. H. Christensen, and C. Scheutz. 2011. “Mass Balances and Life-Cycle Inventory of Home Composting of Organic Waste.” *Waste Management (New York, N.Y.)* 31 (9–10): 1934–1942. doi:10.1016/j.wasman.2011.05.004.
- Barrington, S., D. Choinière, M. Trigui, and W. Knight. 2002. “Effect of Carbon Source on Compost Nitrogen and Carbon Losses.” *Bioresource Technology* 83 (3): 189–194. doi:10.1016/S0960-8524(01)00229-2.
- Bernal, M. P., J. A. Albuquerque, and R. Moral. 2009. “Composting of Animal Manures and Chemical Criteria for Compost Maturity Assessment. A Review.” *Bioresource Technology* 100 (22): 5444–5453. doi:10.1016/j.biortech.2008.11.027.
- Breitenbeck, G. A., and D. Schellinger. 2004. “Calculating the Reduction in Material Mass and Volume during Composting.” *Compost Science & Utilization* 12 (4): 365–371. Published online in 2013. doi:10.1080/1065657X.2004.10702206.
- Brito, L. M., J. Coutinho, and S. R. Smith. 2008. “Methods to Improve the Composting Process of the Solid Fraction of Dairy Cattle Manure.” *Bioresource Technology* 99 (18): 8955–8960. doi:10.1016/j.biortech.2008.05.005.
- Burkholder, J., B. Libra, P. Weyer, S. Heathcote, D. Kolpin, P. S. Thorne, and M. Wichman. 2007. “Impacts of Waste from Concentrated Animal Feeding Operations on Water Quality.” *Environmental Health Perspectives* 115 (2): 308–312. doi:10.1289/ehp.8839.
- Burton, C., and C. Turner. 2003. *Manure Management: Treatment Strategies for Sustainable Agriculture*. Silsoe Research Institute, Silsoe, UK: Silsoe Research Institute.
- CEM Corp. 2006. *CEM-MDS 2000 Microwave Digester*. Matthews, NC: CEM Corporation.
- David, M. B., L. E. Drinkwater, and G. F. McIsaac. 2010. “Sources of Nitrate Yields in the Mississippi River Basin.” *Journal of Environmental Quality* 39 (5): 1657–1667. doi:10.2134/jeq2010.0115.
- Dou, Z., G. Y. Zhang, W. L. Stout, J. D. Toth, and J. D. Ferguson. 2003. “Efficacy of Alum and Coal Combustion by-Products in Stabilizing Manure Phosphorus.” *Journal of Environmental Quality* 32 (4): 1490–1497. doi:10.2134/jeq2003.1490.
- Edwards, D. R., and T. C. Daniel. 1993. “Abstractions and Runoff from Fescue Plots Receiving Poultry Litter and Swine Manure.” *Transactions of the American Society of Agricultural Engineers* 36 (2): 405–411. doi:10.2134/jpa1993.0492.
- Eghball, B., and J. F. Power. 1999. “Composted and Noncomposted Manure Application to Conventional and no-Tillage Systems: Corn Yield and Nitrogen Uptake.” *Agronomy Journal* 91 (5): 819–825. doi:10.2134/agronj1999.915819x.
- Fan, H., J. Liao, O. K. Abass, L. Liu, X. Huang, L. Wei, J. Li, W. Xie, and C. Liu. 2019. “Effects of Compost Characteristics on Nutrient Retention and Simultaneous Pollutant Immobilization and Degradation during co-Composting Process.” *Bioresource Technology* 275: 61–69. doi:10.1016/j.biortech.2018.12.049.
- Farran, T. B., G. E. Erickson, T. J. Klopfenstein, C. N. Macken, and R. U. Lindquist. 2006. “Wet Corn Gluten Feed and Alfalfa Hay Levels in Dry-Rolled Corn Finishing Diets: Effects on Finishing Performance and Feedlot Nitrogen Mass Balance.” *Journal of Animal Science* 84 (5): 1205–1214. doi:10.2527/2006.8451205x.
- Hansen, M. N., K. Henriksen, and S. G. Sommer. 2006. “Observations of Production and Emission of Greenhouse Gases and Ammonia during Storage of Solids Separated from Pig Slurry: Effects of Covering.” *Atmospheric Environment* 40 (22): 4172–4181. doi:10.1016/j.atmosenv.2006.02.013.
- Hansen, N. C., and S. Goyal. 2001. “Runoff Water Quality and Crop Responses to Variable Manure Application Rates.” WRC Research 2001. Morris, MN: West Central Research and Outreach Center, University of Minnesota.
- Haug, R. T. 1993. *The Practical Handbook of Compost Engineering*. Boca Raton, FA: Lewis Publishers and CRC Press.
- Heathwaite, A. L., P. Griffiths, and R. J. Parkinson. 1998. “Nitrogen and Phosphorus in Runoff from Grassland with Buffer Strips following Application of Fertilizers and Manures.” *Soil Use and Management* 14 (3): 142–148. doi:10.1111/j.1475-2743.1998.tb00140.x.
- IEM. 2018. “Iowa Environmental Mesonet.” Iowa State University Website for accessing Iowa climate data records. National Weather Service Cooperative Observer Program. Iowa State University, Ames, IA: Iowa Environmental Mesonet. Accessed January 12, 2018. <https://mesonet.agron.iastate.edu/request/coop/fe.phtml>.
- Jacobson, L. M., M. B. David, and L. E. Drinkwater. 2011. “A Spatial Analysis of Phosphorus in the Mississippi River Basin.” *Journal of Environmental Quality* 40 (3): 931–941. doi:10.2134/jeq2010.0386.
- James, E., P. Kleinman, T. Veith, R. Stedman, and A. Sharpley. 2007. “Phosphorus Contributions from Pastured Dairy Cattle to Streams of the Cannonsville Watershed, New York.” *Journal of Soil and Water Conservation* 62 (1): 40–47.
- Lam, Q. D., B. Schmalz, and N. Fohrer. 2011. “The Impact of Agricultural Best Management Practices on Water Quality in a North German Lowland Catchment.” *Environmental Monitoring and Assessment* 183 (1–4): 351–379. doi:10.1007/s10661-011-1926-9.

- Larney, F. J., A. F. Olson, J. J. Miller, and B. C. Tovell. 2014a. "Nitrogen and Phosphorus in Runoff from Cattle Manure Compost Windrows of Different Maturities." *Journal of Environmental Quality* 43 (2): 671–680. doi:10.2134/jeq2013.06.0230.
- Larney, F. J., A. F. Olson, J. J. Miller, and B. C. Tovell. 2014b. "Soluble Salts, Copper, Zinc, and Solids Constituents in Surface Runoff from Cattle Manure Compost Windrows." *Canadian Journal of Soil Science* 94 (4): 515–527. doi:10.4141/cjss2013-114.
- Maynard, A. 1993. "Nitrate Leaching from Compost-Amended Soils." *Compost Science & Utilization* 1 (2): 65–72. doi:10.1080/1065657X.1993.10757875.
- Parry, R. 1998. "Agricultural Phosphorus and Water Quality: A US Environmental Protection Agency Perspective." *Journal of Environmental Quality* 27 (2): 258–261. doi:10.2134/jeq1998.00472425002700020003x.
- Perkin-Elmer Inc. 2004. *P2400 Elemental Analyzer and AANALYST-100 Atomic absorption Spectrophotometer*. Waltham, MA: Perkin-Elmer, Inc.
- Ramos, M. C., J. N. Quinton, and S. F. Tyrrel. 2006. "Effects of Cattle Manure on Erosion Rates and Runoff Water Pollution by Faecal Coliforms." *Journal of Environmental Management* 78 (1): 97–101. doi:10.1016/j.jenvman.2005.04.010.
- Robinson, R., E. Kimmel, and Y. Avnimelech. 2000. "Energy and Mass Balances of Windrow Composting System." *Transactions of the American Society of Agricultural and Biosystems Engineers* 43 (5): 1253–1259. doi:10.13031/2013.3019.
- Rynk, R., M. van de Kamp, G. B. Willson, M. E. Singley, T. L. Richard, J. J. Kolega, F. R. Gouin, Jr., et al. 1992. *On-Farm Composting Handbook*. Ithaca, NY: NRAES, Cornell University.
- SAS Institute. 2004. *SAS Software and User's Guide*. Cary, NC: SAS Institute.
- Sefeedpari, P., T. Vellinga, S. Rafiee, M. Sharifi, P. Shine, and S. H. Pishgar-Komleh. 2019. "Technical, Environmental and Cost-Benefit Assessment of Manure Management Chain: A Case Study of Large Scale Dairy Farming." *Journal of Cleaner Production* 233: 857–868. doi:10.1016/j.jclepro.2019.06.146.
- Seymour, R. M., and M. Bourdon. 2003. "Hydrology and Nutrient Movement of a Windrow of Dairy Bedding/Leaf Mulch Compost." *Paper presented at the 2003 American Society of Agricultural and Biological Engineers Annual International Meeting, Riviera Hotel and Convention Center, Las Vegas, Nevada, USA, 27–30 July 2003*. ASABE Technical Library.
- Sikora, L. J., and H. Francis. 2000. "Lime-Stabilized Soil for Use as a Compost Pad." U.S. Department of Agriculture-Agricultural Research Service Beltsville, MD: U.S. Department of Agriculture.
- Smith, D. R., P. R. Owens, A. B. Leytem, and E. A. Warnemuende. 2007. "Nutrient Losses from Manure and Fertilizer Applications as Impacted by Time to First Runoff Event." *Environmental Pollution (Barking, Essex: 1987)* 147 (1): 131–137. doi:10.1016/j.envpol.2006.08.021.
- Tiquia, S. M., T. L. Richard, and M. S. Honeyman. 2002. "Carbon, Nutrient and Mass Loss during Composting." *Nutrient Cycling in Agricultural Ecosystems* 62 (1): 15–24.
- USEPA. 2009. *National Water Quality Inventory: Report to Congress, 2004 Reporting Cycle*. EPA 841-R-08-001. Washington, D.C.: U.S. Environmental Protection Agency.
- USGS. 2010. *The Quality of Our Nation's Water: Nutrients in the Nation's Streams and Groundwater, 1992–2004*. USGS Circular 1350. Reston, Va.: U.S. Geological Survey.
- Vervoort, R. W., D. E. Radcliffe, M. L. Cabrera, and M. Latimore, Jr. 1998. "Field-Scale Nitrogen and Phosphorus Losses from Hayfields Receiving Fresh and Composted Broiler Litter." *Journal of Environmental Quality* 27 (5): 1246–1254. doi:10.2134/jeq1998.004724250027000500033x.
- Webber, D. F., S. K. Mickelson, T. L. Richard, and H. K. Ahn. 2009. "Effects of a Livestock Manure Windrow Composting Site with a Fly Ash Pad Surface and Vegetative Buffers on Sediment, Nitrate, and Phosphorus Losses with Runoff." *Journal of Soil and Water Conservation* 64 (2): 163–171. doi:10.2489/jswc.64.2.163.
- Webber, D., S. K. Mickelson, T. L. Richard, and H. K. Ahn. 2016. "Application of a Vegetative Buffer and Fly Ash Pad Surface Material System for Reducing Runoff Flow Volume, Sediment, and Nutrient Losses from Livestock Manure Windrow Composting Facilities." In Vol. 47 of *Advances in Environmental Research*, edited by J.A. Daniels, 39–59. New York: Nova Science Publishers.
- Wilson, B. G., K. Haralampides, and S. Levesque. 2004. "Stormwater Runoff from Open Windrow Composting Facilities." *Journal of Environmental Engineering and Science* 3 (6): 537–540. doi:10.1139/s04-024.